MLP: neuron displacement caps and matched-norm controls

All gradients and Adam moments retained; first-layer updates transformed from 448; all learning rates stay 0.001.

Unchanged Adam — Selective policy (column color) — Uniform control: matched first-layer norm

I17: Neuron cap: prefix q90
Fixed threshold 0.15155

I18: Neuron cap: prefix q75
Fixed threshold 0.098915

I19: Neuron cap: prefix q50
Fixed threshold 0.052223

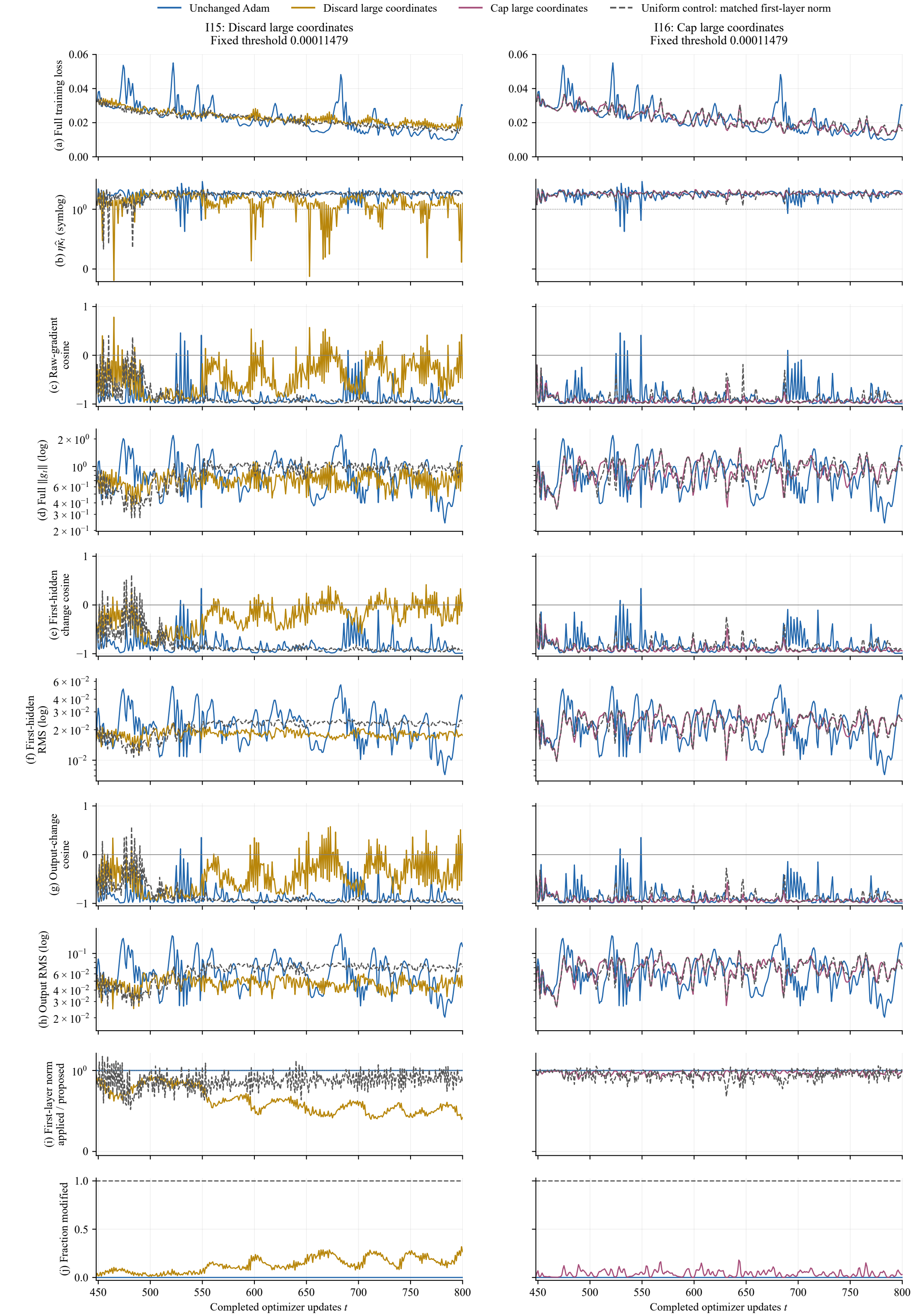
(a) Full training loss
(b) $\hat{\eta}\hat{\kappa}_t$ (symlog)
(c) Raw-gradient cosine
(d) Full $\|g_t\|$ (log)
(e) First-hidden change cosine
(f) First-hidden RMS (log)
(g) Output-change cosine
(h) Output RMS (log)
(i) First-layer norm applied / proposed
(j) Fraction modified

Completed optimizer updates t

Loss: independent linear axes. Other rows use common scales. All 5,000 samples; outgoing changes; no smoothing.
Row (i): first-layer norm after modification / native proposal; a uniform control may exceed 1. Row (j): coordinate or neuron fraction modified.
Thresholds use only untreated updates 416-447. Norm controls follow the selective arm at the same update; only first-layer norms are matched.

MLP: coordinate gates and matched-norm controls

All gradients and Adam moments retained; first-layer updates transformed from 448; all learning rates stay 0.001.



Loss: independent linear axes. Other rows use common scales. All 5,000 samples; outgoing changes; no smoothing.
Row (i): first-layer norm after modification / native proposal; a uniform control may exceed 1. Row (j): coordinate or neuron fraction modified.
Thresholds use only untreated updates 416-447. Norm controls follow the selective arm at the same update; only first-layer norms are matched.